FitLife Studio Analytics

Business Plan for Membership Renewal Risk Prediction

Prepared for: Decision Tree Model and Business Interpretation Assignment

Business area	Fitness subscription services and customer retention
Business decision	Identify members who are at risk of not renewing their membership
Machine learning task	Binary classification using a Decision Tree model
Target variable	At_Risk_Not_Renewing
Positive class	Yes = member is predicted to be at risk of not renewing

1. Business Overview

FitLife Studio is a mid-sized fitness studio chain that sells monthly, six-month, and annual memberships. The business offers gym access, group classes, personal training, and a mobile app that helps members book classes and track visits. The company earns most of its revenue from recurring membership renewals, so keeping existing members is a major business priority.

The studio operates in a competitive market where customers can easily switch to another gym, home workout app, or lower-cost fitness option. Because of this, FitLife wants to use data to identify early signs that a member may not renew.

Business goal: Use customer engagement and service data to support timely retention actions before the renewal date.

2. Business Problem

The main problem is membership churn. Some members stop attending classes, log into the app less often, report lower satisfaction, or experience payment and service problems before they decide not to renew. If the business waits until the membership expires, it may be too late to recover the customer relationship.

Business challenge	Why it matters	Possible business action
Low member engagement	Members who rarely visit may feel less value from the membership.	Send personalized class suggestions or check-in messages.
Low satisfaction or complaints	Dissatisfied members are more likely to leave.	Assign service follow-up or offer a recovery plan.
Payment issues	Payment problems can create friction and cancellation risk.	Offer billing support or flexible payment options.
High distance from studio	Convenience affects renewal decisions.	Recommend online classes or closer branch options.
Monthly contract type	Monthly members can leave more easily than annual members.	Offer renewal incentives or longer contract benefits.

3. Machine Learning Decision

A Decision Tree classification model can help FitLife predict whether a member is at risk of not renewing. The prediction can help managers decide which members should receive a retention intervention, such as a reminder, personalized promotion, class recommendation, or service follow-up.

Decision supported by the model: Should FitLife proactively contact this member with a retention action before the renewal period ends?

Target variable: At_Risk_Not_Renewing. The value Yes means the member is considered at risk of not renewing. The value No means the member is not currently considered at high renewal risk.

4. Dataset Description

The provided Excel dataset contains synthetic member-level records. Each row represents one FitLife member. The dataset includes demographic information, membership details, engagement indicators, satisfaction indicators, and the target variable.

Feature group	Example columns	Business meaning	Expected relationship to risk
Membership details	Membership_Type, Monthly_Fee, Contract_Type, Months_As_Member	Shows the customer relationship and contract structure.	Monthly contracts and shorter relationships may increase risk.
Engagement	Avg_Weekly_Visits, Classes_Per_Month, App_Login_Days_Last_30	Shows how actively the member uses the service.	Lower engagement may increase risk.
Service experience	Satisfaction_Score, Complaint_Count_Last_3_Months	Shows whether the member is satisfied with the experience.	Lower satisfaction and more complaints may increase risk.
Payment and renewal	Payment_Issues_Last_6_Months, Renewal_Reminder_Sent, Last_Renewal_Discount_Used	Shows potential renewal friction and retention communication.	Payment issues may increase risk; reminders may reduce risk.
Context	Distance_From_Studio_km, Corporate_Plan, Season, Referral_Source	Shows external and contextual factors that may influence renewal.	Longer distance and seasonal patterns may affect risk.

Important note for students: The dataset intentionally includes a small number of missing values and duplicate rows so that data inspection and cleaning can be practiced before model training.

5. How the Model Results Should Be Interpreted

Students should not only report model scores. They should explain what the results mean for the business. For example, a high testing accuracy may suggest that the model is useful, but the business should also consider recall, precision, and the meaning of errors.

Evaluation concept	Business interpretation	Why it matters
Accuracy	The percentage of correct predictions across both risk and non-risk members.	Useful as a general score, but it may hide problems if classes are imbalanced.
Precision for Yes	Of the members predicted as at risk, how many were actually at risk.	Important when retention resources are limited and the business wants fewer wasted offers.
Recall for Yes	Of all actually at-risk members, how many the model correctly identified.	Very important because missing at-risk members can lead to lost renewals and lost revenue.
False positive	The model flags a member as at risk, but the member would likely renew anyway.	The business may waste a discount or staff time.
False negative	The model predicts the member is not at risk, but the member is actually at risk.	The business may fail to contact a member who needs support, causing preventable churn.
Overfitting check	Compare training accuracy and testing accuracy.	If training accuracy is much higher than testing accuracy, the Decision Tree may not generalize well.

6. Business Use of Important Features

The Decision Tree may identify features such as satisfaction score, average weekly visits, app login days, payment issues, complaints, and contract type as important. These features can help FitLife understand why some members are more likely to leave and where the business should intervene.

Examples of possible actions based on important features:

- If low weekly visits are important, FitLife can recommend beginner-friendly classes or send motivational check-ins.
- If low satisfaction is important, managers can prioritize service recovery conversations.
- If payment issues are important, billing support can contact the member before cancellation occurs.
- If monthly contracts are important, the business can design annual membership incentives.
- If distance is important, the business can promote online classes or suggest a more convenient location.

7. Limitations, Bias, and Responsible Use

The model should support business judgment, not replace it. The dataset is synthetic and may not capture all reasons why a person renews or cancels a membership. Real customer decisions may be influenced by health, income, schedule changes, family responsibilities, transportation, or personal preferences that are not fully represented in the data.

There is also a risk of bias if the business treats some groups differently based only on model output. For example, if the model overpredicts risk for certain membership types or locations, those members may receive different offers or more frequent contact. Managers should review model outputs carefully and avoid using the model in a way that creates unfair treatment.

Responsible AI reflection: Human judgment should still be used because the model cannot fully understand customer context, fairness concerns, or long-term brand relationships.

8. Assignment Alignment

This business plan is designed to help students connect a Decision Tree classification model to a realistic business decision. Students should use the dataset to complete data preparation, model training, evaluation, overfitting analysis, feature importance interpretation, and business interpretation.

Assignment question	Expected connection to this business plan
What is the business about?	Fitness membership services and customer retention.
What problem is the business trying to solve?	Predict and reduce membership non-renewal risk.
What decision can ML help make?	Which members should receive proactive retention support.
What is the target variable?	At_Risk_Not_Renewing.
What metric is most important?	Recall for the Yes class may be most important if missing at-risk members is more costly.
Why use human judgment?	To review context, avoid unfair treatment, and decide appropriate retention actions.